

TUBERCULOSIS

Tuberculosis is a chronic infectious disease
caused by *Mycobacterium*
tuberculosis

- TB in children is mainly due to failure of TB
control in adults

ETIOLOGY

- M. tuberculosis complex: - M. tuberculosis, (the most important cause in
- humans), M. bovis, M. africanum, M. microti and M. Canetti

EPIDEMIOLOGY

- .The World Health Organization (WHO) estimates that since 2015, tuberculosis
- .has surpassed (HIV/AIDS) as the leading cause of death from an infectious disease worldwide

MODE OF TRANSMISSION

- Inhalation: - pulmonary tuberculosis
- Infectious droplet nuclei can be spread into the air by:
 - Coughing (single coughing Can produce 3000 droplet nuclei)
 - Talking
 - Sneezing
 - Spitting and singing

CLINICAL MANIFESTATIONS

- .Cough of greater than 2 week
- .Night sweating
- .Loss of appetite
- .Weight loss
- .Recurrent pulmonary infection with no improvement with antibiotics
- .Contact with chronic cougher or tb diagnosed

Classification

.1.PULMONARY TUBERCULOSIS

.2.. EXTRA PULMONARY TUBERCULOSIS

PULMONARY TUBERCULOSIS

- .it refers to any bacteriologically confirmed or clinically diagnosed case of TB
- .involving the lung parenchyma or the tracheobronchial tree

EXTRA PULMONARY TUBERCULOSIS

- refers to any bacteriologically confirmed or clinically diagnosed case of TB
- involving organs other than the lungs, e.g. pleura, lymph nodes, abdomen,
- genitourinary tract, skin, joints and bones, meninges

TYPES OF EXTRA-PULMONARY TUBERCULOSIS ARE

- .TUBERCULOUS LYMPHADENOPATHY
- . PLEURAL TB
- .PERICARDIAL TB
- .TB OF CNS
- .TUBERCULOMA
- .BONE AND JOINT TB

.SPINAL TUBERCULOSIS

.ABDOMINAL AND GASTROINTESTINAL
DISEASE

.TUBERCULOSIS ENTERITIS

.TUBERCULOUS PERITONITIS

.GENITAL TRACT TUBERCULOSIS

MOST COMMON

.TUBERCULOUS LYMPHADENOPATHY

- Clinical picture

- TB of Superficial LNs (Scrofula) is most common form of EPTB Common sites are cervical lymph node, occasionally axillary and intraabdominal LN may be involved Initially cervical lymph nodes are discrete, firm and non-tender may

- later be matted together and become fluctuant

- .It is usually diagnosed by (FNA) of the lymph node
- . A definitive diagnosis of tuberculous adenitis usually requires:histologic or bacteriologic confirmation, which is best accomplished by FNA for culture, stain, and histology Culture of lymph node tissue yields the organism in only
- .approximately 50% of cases

TB MENINGITIS

.THE 3 STAGES OF TB MENINGITIS

.1st stage

- .o lasts 1–2 weeks

- .o characterized by nonspecific symptoms (fever, headache, irritability, drowsiness, and malaise)

- .o Focal neurologic signs are absent

.2nd stage

.o usually begins more abruptly

.o The most common features are:

.▪ Lethargy, nuchal rigidity, seizures positive Kernig or Brudzinski

.signs, hypertonia

.▪ vomiting, cranial nerve palsies, and other focal

- 3rd stage

- . it is marked by coma, hemiplegia or paraplegia, hypertension,
- .decerebrate posturing, deterioration of vital signs, and eventually
- .death.

Diagnosis

.The CSF Analysis

- . Leukocyte count -10 to 500 cells/mm³
- . Glucose <40 mg/dL
- . Protein - 400–5,000 mg/dL
- . AFB and culture yield depends on volume (if 5-10ml: AFB is
- .positive in 30%, culture positive in 50-70%)

INVESTIGATIONS (DIAGNOSTIC TOOLS)

.1..BACTERIOLOGICAL METHODS

.MICROSCOPIC EXAMINATION

- The most specific confirmation is isolation of *M.tuberculosis*
- Spot sputum smear examination with two sample; the second taken 30-60 minutes after the first one
- Early morning gastric aspirate (young age)

.2. MOLECULAR TEST

.Gene Xpert

- .o It is a new and rapid DNA test for TB and it can detect Rifampin resistance currently preferred first line diagnostic test

- . Indicated for diagnosis of TB in high MDR-TB and TB/HIV

- .o For diagnosis of TB in children and Extra

3. HISTO-PATHOLOGICAL EXAMINATION

.PLEURAL FLUID ANALYSIS

- .Color: Yellow with blood tinged
- Chemical: Protein (2-4gm/dl,
- glucose (20-40 mg/dl)
- Cells: lymphocyte predominance

4. RADIOLOGIC EXAMINATION

.CHEST X-RAY

- Upper lobe infiltration
- Enlarged hilar and mediastinal lymph nodes and opacification in lung tissue
- Cavitation (common with older children), pleural effusion, emphysema,
- .collapse

.X-RAY OF BONES

- Narrowing of disc space
- Collapse of vertebral body, Extensive destruction with kyphosis (pott disease)

.CT/MRI

- Diffuse brain edema and lepto-meningeal inflammation, meningeal enhancement (TB meningitis)
- Discrete lesions with significant surrounding edema and ring enhancement (Tuberculoma)

TUBERCULOSIS TREATMENT PRINCIPLES

- .Antituberculosis drugs
- .Nutritional rehabilitation
- .Screening of family members and and/or other suspected index cases: for
- .prevention and treatment
- Follow up for (Adherence, response to treatment, drug side effect)

Antituberculosis drugs

- .Ethambutol
- .Isoniazid
- .Pyrazinamide
- .Rifampin

PHASES OF CHEMOTHERAPY AND DAILY DOSING FREQUENCY

.INTENSIVE PHASE

- Reduces bacterial load(makes patient noninfectious)
- Prevent emergence of drug resistant strains
- Treatment with two months of RHZE

.CONTINUATION PHASE

- To ensure the patient is permanently cured and no relapse after completion of
- .treatment
- To clear dormant bacilli
- Treatment with Four months of RH (exceptions